Density Formula Triangle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS The triangle links mass (m), density (d), and volume (v). Cover the unknown to read the formula: $m = d \times v$, $d = m \div v$, and $v = m \div d$. Here d means density, not distance.

Round 1 — Retrieve It

1. What three quantities does the density triangle connect?
2. Write the formula for mass.
3. Write the formula for density.
4. Which variable do you cover to find volume, and what formula results?

Round 2 — Sketch It

Draw the density triangle. Put mass on top; put density and volume on the bottom. Circle the two quantities you would multiply to find mass.

Density — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. A block has a density of 4 g/cm^3 and a volume of 5 cm^3. What is its mass?
2. A sample has a mass of 60 g and a volume of 12 cm^3. What is its density?
3. A metal cube has a mass of 48 g and a density of 6 g/cm^3. What is its volume?

Round 4 — Reason from Evidence

Two objects each have a volume of 10 cm^3. Object A has a mass of 30 g; Object B has a mass of 20 g. Use the triangle to find each object's density, then say which object is denser and how you know.

Density — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A Articulate

Explain how the density triangle helps you find a missing quantity. Use one example that is not printed on the cue card.

C Connect

Connect density to whether an object floats or sinks in water. Water has a density of 1 g/cm^3. What does a density greater than or less than 1 g/cm^3 tell you?

E Extend

Create your own problem that asks for volume. Give the numbers and show the work using $v = m \div d$.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Density

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Mass, density, and volume.
2. $m = d \times v$.
3. $d = m \div v$.
4. Cover v; $v = m \div d$.

Sketch It — look for

Mass on top, with density and volume on the bottom. Mass is found by multiplying density × volume (the two bottom quantities).

Apply It

1. $m = d \times v = 4 \times 5 =$ **20 g**.
2. $d = m \div v = 60 \div 12 =$ **5 g/cm³**.
3. $v = m \div d = 48 \div 6 =$ **8 cm³**.

Reason from Evidence — look for

Object A: $d = 30 \div 10 = 3$ g/cm³. Object B: $d = 20 \div 10 = 2$ g/cm³. Object A is denser because it has more mass in the same volume.

ACE — Extend look-for

Many answers possible. Check that the problem asks for volume and the work correctly uses $v = m \div d$.

> **Suggested use.** Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.